Weekly Progress Report

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Domain: Full Stack Development
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Week Ending: 02

I. Overview:

This week focused on developing the core functionality of the CMS Blog project by implementing the Blog Post system. The primary objective was to build REST APIs supporting CRUD (Create, Read, Update, Delete) operations and integrate them with the MySQL database using Spring Boot

II. Achievements:

1. Blog Post Module Development:
   - Created Post entity with fields such as title, content, author, and createdAt
   - Used JPA annotations for ORM mapping
   - Configured table creation in MySQL
2. Repository Layer:
   - Developed PostRepository using Spring Data JPA
   - Enabled database operations without writing SQL queries
3. Service Layer:
   - Implemented PostService to handle business logic
   - Developed methods for:
     - Creating posts
     - Retrieving all posts
     - Fetching post by ID
     - Updating posts
     - Deleting posts.

4. REST API Development:
   - Created PostController to expose REST endpoints
   - Implemented APIs:
     - POST /posts → Create post
     - GET /posts → Fetch all posts
     - GET /posts/{id} → Fetch single post
     - PUT /posts/{id} → Update post
     - DELETE /posts/{id} → Delete post

5.Database Integration:
   - Successfully connected Spring Boot application with MySQL
   - Verified automatic table creation and data persistence

6.API Testing:
   - Tested all endpoints using Postman
   - Verified multiple test cases including:
     - Creating multiple posts
     - Updating and deleting records
     - Handling invalid inputs

7.GitHub Updates:
   - Committed and pushed Week 2 code to GitHub
   - Resolved merge conflicts and successfully updated repository

III. Challenges:

- Faced Maven command issues and resolved using Maven Wrapper (mvnw.cmd)
- Encountered package mismatch error due to incorrect folder structure and fixed it
- Faced API 404 error due to incorrect endpoint and resolved it
- Handled Git push rejection and merge conflicts during repository update

IV. Learning Outcomes:

- Gained strong understanding of CRUD operations in REST APIs
- Learned layered architecture (Controller, Service, Repository) in Spring Boot
- Improved debugging and problem-solving skills
- Gained experience in API testing and backend development

V. Next Week's Goals:

- Develop frontend using React
- Create UI for blog creation and display
- Integrate frontend with backend APIs
- Enhance user experience with proper UI design

VI. Additional Comments:

Week 2 was highly productive as it involved building the core functionality of the CMS project. Successfully implemented and tested the Blog Post system, which forms the backbone of the application. Confidence has improved in backend development and API design..